FORM 2

THE PATENTS ACT, 1970

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

SYSTEM FOR MONITORING ACTIVE SAFETY OF A VEHICLE AND METHOD THEREOF

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure relates to the field of vehicular safety. In particular, the present disclosure related to a system and method for monitoring active safety of a vehicle.

BACKGROUND

[0002] The following description of the related art is intended to provide background information pertaining to the field of the disclosure. This section may include certain aspects of the art that may be related to various features of the present disclosure. However, it should be appreciated that this section is used only to enhance the understanding of the reader with respect to the present disclosure, and not as admission of the prior art.

[0003] Land transport is undeniably most crucial mode of transport due to high flexibility, greater reach, and low-cost factor, and is used to move people, livestock, and cargo. Given the volume of passengers and cargo that is moved via land vehicles, safety of passengers and property being moved via land vehicles is a prime challenge. Given the variety of terrains through which land vehicles are driven, vehicle safety becomes many folds crucial. The safety of passengers and property is a serious concern especially in the case of mountainous terrains where the roads have steep inclinations, and sharp narrow bends.

[0004] The present approaches to vehicular safety provide passive safety features designed to protect occupants in the event of a crash, like seatbelts, airbags, and crumple zones. These safety approaches are far from being sufficient when the risk of driving through a terrain is very high.

[0005] Therefore, there is a need for solutions that provide active vehicular safety to prevent crashes.

OBJECTS OF THE PRESENT DISCLOSURE

[0006] Some of the objects of the present disclosure, which at least one embodiment herein satisfies are as listed below.

[0007] One object of the present disclosure is to provide a system and method for monitoring active safety of a vehicle.

[0008] Another object of the present disclosure is to provide a system and a method for monitoring active safety of a vehicle which can also be used to assess a driver in terms of his/her driving skills, actions, and practices for safety, and conditions associated with a vehicle in real-time and/or over a period of time.

[0009] Another object of the present disclosure is to provide a system and a method for monitoring active safety of a vehicle that can be used in investigating safety violations, and performing analytics for developing interventions to promote vehicular safety by regulatory agencies and authorities.

SUMMARY

[0010] The present disclosure relates to the field of vehicular safety. In particular, the present disclosure related to a system and method for monitoring active safety of a vehicle.

[0011] In an aspect, the present disclosure relates to a system for active vehicular safety. The system includes one or more first image detection units configured with an interior of a vehicle to detect one or more first images of the interior of the vehicle, where the one or more first image detection units are communicatively coupled to the system. The system includes one or more second image detection units configured with an exterior of a vehicle to detect one or more second images of one or more surroundings of the vehicle, where the one or more second image detection units are communicatively coupled to the system. The system includes one or more sensors configured with the vehicle to detect one or more parameters associated with the vehicle and a movement of the vehicle to monitor for a vehicular safety, where the one or more sensors are communicatively coupled to the system. The system includes at least one processor, and a memory operatively coupled with the processor, where said memory stores executable instructions which when executed by the processor, the processor performs the following steps. The processor receives, data comprising one or more first images, one or more second images, and one or more parameters, via one or more first image

detection units, one or more second image detection units, and one or more sensors, at one or more predetermined periods by a data collection engine. The processor detects, by a detecting and predicting engine, one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, reduction of load in the vehicle, distribution of load, and the like. The processor may analyze, the data, using an Artificial Intelligence (AI) engine. The processor may detect one or more risk factors, based on the analysis of the data by a processor. The processor alerts, by an alerting and reporting engine, secondary user(s) regarding risk factors, traffic violations, and the like.

[0012] In an embodiment, the one or more sensors may include one or more pressure sensors to detect pressure associated with a plurality of wheels of the vehicle, one or more luggage compartments, and one or more rooftop luggage carriers, indicative of the weight distribution across the vehicle.

[0013] In an embodiment, the processor may analyze the one or more parameters to determine a total load configured with the vehicle, and a distribution of the total load with one or more locations of the vehicle.

[0014] In an embodiment, the processor may analyze one or more first images for determining a number of passengers within the vehicle.

[0015] In an embodiment, the speed of the vehicle is determined by the processor embedded with an artificial intelligence engine using the rate of change of the video frames captured.

[0016] In an embodiment, the processor may notify one or more secondary users via one or more computing devices associated with the one or more secondary users about one or more safety violations, based on the analysis of the data.

[0017] In an embodiment, the processor stores the data with one or more first images, the one or more second images, and one or more parameters within a database.

[0018] In an embodiment, the processor suggests an optimum speed to a primary user at the one or more predetermined periods, based on the analysis of the data.

[0019] In an embodiment, the processor evaluates and quantifies a primary user for the vehicular safety, based on the analysis of the data.

[0020] In an embodiment, the system includes a Blackbox housing to record all data obtained from the image detection units and pressure sensors. The driver's overall safety performance is accessed, with the findings stored in the Blackbox housing for future retrieval and examination and the secondary user have access to the Blackbox data for the identification of any safety breaches.

[0021] In an embodiment, the system is entirely self-sufficient, including its own battery supply, and not connected to any existing sub-system of the vehicle to prevent tampering.

[0022] In an aspect, the present disclosure relates to a method for monitoring active safety of a vehicle. The method includes the following steps. Receiving, by at least one processor associated with a system, data comprising one or more first images, one or more second images, and one or more parameters, via one or more first image detection units, one or more second image detection units, and one or more sensors, at one or more predetermined periods. Detecting, by a detecting and predicting engine, one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, driver responses, reduction of load in the vehicle, distribution of load, and the like. Analyzing, by the processor, the data, using an Artificial Intelligence (AI) engine. Recording, by a Blackbox housing, data received from the data collection engine and accessing the overall safety performance of a primary user based on the findings stored in the Blackbox housing for future retrieval and examination; wherein the secondary user have access to the Blackbox data for the identification of any safety breaches. Detecting, by the processor, one or more risk factors, based on the analysis of the data from the Blackbox housing. Alerting, by an alerting and reporting engine, secondary user(s) regarding risk factors, traffic violations, and the like.

BRIEF DESCRIPTION F DRAWINGS

[0023] The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in, and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure, and together with the description, serve to explain the principles of the present disclosure.

[0024] In the figures, similar components, and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0025] FIG. 1 illustrates (100) an example system (108) for monitoring active safety of a vehicle, in accordance with an embodiment of the present disclosure.

[0026] FIG. 2 illustrates (200) a block diagram of an example system (108) similar to the one shown in FIG.1 with functional components, in accordance with an embodiment of the present disclosure.

[0027] FIG. 3 illustrates an example method (300) for monitoring active safety of a vehicle, in accordance with an embodiment of the present disclosure.

[0028] FIG. 4 illustrates an example computer system (400) in the embodiments of the present disclosure.

DETAILED DESCRIPTION

[0029] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

[0030] The ensuing description provides exemplary embodiments only, and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing an exemplary embodiment. It should be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention as set forth.

[0031] Specific details are given in the following description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, circuits, systems, networks, processes, and other components may be shown as components in block diagram form in order not to obscure the embodiments in unnecessary detail. In other instances, well-known circuits, processes, algorithms, structures, and techniques may be shown without unnecessary detail in order to avoid obscuring the embodiments.

[0032] Also, it is noted that individual embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed but could have additional steps not included in a figure. A process may correspond to method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination can correspond to a return of the function to the calling function or the main function.

[0033] The word “exemplary” and/or “demonstrative” is used herein to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques

known to those of ordinary skill in the art. Furthermore, to the extent that the terms "includes," "has," "contains," and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive—in a manner similar to the term "comprising" as an open transition word—without precluding any additional or other elements.

[0034] Reference throughout this specification to "one embodiment" or "an embodiment" or "an instance" or "one instance" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0035] The various aspects of the present disclosure have been described with reference to FIGs. 1-4.

[0036] The present disclosure relates to the field of vehicular safety. In particular, the present disclosure related to a system and method for monitoring active safety of a vehicle.

[0037] In an aspect, the present disclosure relates to a system for monitoring active safety of a vehicle. Active vehicular safety refers to the safety measures/mechanisms in place to prevent vehicular accidents and crashes instead of passive measures/mechanisms which reduce the impact and damage of a crash/accident. FIG. 1 illustrates (100) an example system (108) for monitoring active safety of a vehicle, in accordance with an embodiment of the present disclosure. Referring to FIG. 1, the system (108) may include one or more first image detection units configured with an interior of a vehicle to detect one or more first images of the interior of the vehicle, where the one or more first image detection units may be communicatively coupled to the system (108). The system (108) may include one or more second image detection units configured with an exterior of a vehicle to detect one or more second images of one or more surroundings of the vehicle, where the one or more second image detection units

may be communicatively coupled to the system (108). Both first image detection units and second image detection units have been referred to with the numeral 112. The system (108) may include one or more sensors (114) configured with the vehicle to detect one or more parameters associated with the vehicle and a movement of the vehicle to monitor for a vehicular safety, where the one or more sensors (114) may be communicatively coupled to the system (108). In an embodiment, the one or more sensors (114) may include one or more pressure sensors to detect pressure associated with a plurality of wheels of the vehicle, one or more luggage compartments, and one or more rooftop luggage carriers. In an exemplary embodiment, the system (108) may be communicatively coupled to a primary user (110) who may be a vehicle operator or a driver via a monitor just near the dashboard below line of sight of the road. The system (108) may be communicatively coupled to one or more secondary users (102) via one or more computing devices (104) they operate via network (106). The secondary users (102) may include passenger(s), authorities who need to be made known about traffic violations, a transport company, and vehicle owners.

[0038] In an embodiment, the system (108) may perform various steps. The system (108) may receive, data including the one or more first images, the one or more second images, and the one or more parameters, via the one or more first image detection units (112), the one or more second image detection units (112), and the one or more sensors (114), at one or more predetermined periods. In an exemplary embodiment, the predetermined periods (temporal domain) may range between a couple of seconds to a couple of minutes. The system (108) may analyze, the data, using an Artificial Intelligence (AI) engine. The system tracks number of overtakes, classifies the overtakes as safe/unsafe, checks on how many curves, recommended speed maintained by the driver or any combination thereof. The system (108) may detect, one or more risk factors, based on the analysis of the data. In an embodiment, the system (108) may analyze the one or more parameters to determine a total load configured with the vehicle, and a distribution of the total load with one or more locations of the vehicle. In an embodiment, the system (108) may analyze one or more first images for determining a number of passengers

within the vehicle. In an exemplary embodiment, the system (108) may determine the overall balance in the vehicle with respect to number of passengers travelling in the vehicle, load (weight) in terms of luggage and passengers in the vehicle, and distribution of the load across the vehicle in terms of safety requirements.

[0039] In an embodiment, the system (108) may notify the one or more secondary users (102) via one or more computing devices (104) associated with the one or more secondary users (102) about one or more safety violations, based on the analysis of the data. In an embodiment, the system (108) may store the data with one or more first images, the one or more second images, and one or more parameters within a database.

[0040] In an embodiment, the system (108) may suggest an optimum speed to a primary user (110) (driver) at the one or more predetermined periods, based on the analysis of the data. In an exemplary embodiment, the suggestion of optimum speed may be at real-time based on the continuous analysis of various factors including the road conditions, turns, an inclination of the road, vehicular traffic, and a road width. In an embodiment, the system (108) may evaluate and quantify the primary user (110) who is the vehicle operator for the vehicular safety, based on the analysis of the data. In an exemplary embodiment, this evaluation and quantification may be done at real-time and/or over a period of time. In an exemplary embodiment, the system (108) may generate advanced analytics based on individual drivers, vehicles and real-world interventions to promote vehicle safety that may be used by the regulatory agencies/authorities for implementing better regulations, effective safety measures on roads and in vehicles.

[0041] FIG. 2 illustrates (200) a block diagram of an example system (108) for monitoring active safety of a vehicle, similar to the system (108) shown in FIG. 1 with functional components, in accordance with an embodiment of the present disclosure. Referring to FIG. 2, the system (108) may include at least one processor (202) that may be implemented as one or more microprocessors, microcomputers, microcontrollers, digital signal processors, central processing units, logic circuitries, and/or any devices that process data based on operational instructions. Among other capabilities, the processor(s) (202) may be configured to fetch and

execute computer-readable instructions stored in a memory (204) of the system (108). The memory (204) may be configured to store one or more computer-readable instructions or routines in a non-transitory computer readable storage medium, which may be fetched and executed to create or share data packets over a network service. The memory (204) may include any non-transitory storage device including, for example, volatile memory such as random-access memory (RAM), or non-volatile memory such as erasable programmable read only memory (EPROM), flash memory, and the like.

[0042] In an embodiment, the system (108) may include an interface(s) (206). The interface(s) (206) may include a variety of interfaces, for example, interfaces for data input and output devices, referred to as I/O devices, storage devices, and the like. The interface(s) (206) may facilitate communication to/from the system (108). The interface(s) (206) may also provide a communication pathway for one or more components of the system (108). Examples of such components include, but are not limited to, a processing unit/engine(s) (208) and a local database (220). In an embodiment, the local database (220) may be separate from the system (108).

[0043] In an embodiment, the processing engine(s) (208) may be implemented as a combination of hardware and programming (for example, programmable instructions) to implement one or more functionalities of the processing engine(s) (208). In examples described herein, such combinations of hardware and programming may be implemented in several different ways. For example, the programming for the processing engine(s) (208) may be processor-executable instructions stored on a non-transitory machine-readable storage medium and the hardware for the processing engine(s) (208) may include a processing resource (for example, one or more processors), to execute such instructions. In the present examples, the machine-readable storage medium may store instructions that, when executed by the processing resource, implement the processing engine(s) (208). In such examples, the system (108) may include the machine-readable storage medium storing the instructions and the processing resource to execute the instructions, or the machine-readable storage medium may

be separate but accessible to the system (108) and the processing resource. In other examples, the processing engine(s) (208) may be implemented by electronic circuitry.

[0044] In an embodiment, the processing engine (208) may include a data parameter engine (210), an Artificial Intelligence Engine (212), a detecting and predicting engine (214), an alerting and reporting engine (216), and other modules (218). The processor(s) (202) via data parameter engine (210) may receive data from images from image detection units (112), and sensors(s) (114). The processor(s) (202) via the AI engine (212) may analyze the data. The processor(s) (202) via the detecting and predicting engine (214) may detect one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, reduction of load in the vehicle, distribution of load, and the like. The processor(s) (202) via the alerting and reporting engine (216) may alert secondary user(s) regarding risk factors, traffic violations, and the like. Other modules (220) may implement the other functionalities of the system (108) like battery unit, network module etc. In an exemplary embodiment, the AI engine (212) may be trained on a set of roads or paths for driving based on road images, road conditions, and weather conditions such that the pre-trained models may be used for navigation and supplementary support, navigation to the driver to improve vehicular safety. The system (108) includes a Blackbox housing to record all data obtained from the image detection units (112) and pressure sensors, wherein the driver's overall safety performance is accessed, with the findings stored in the Blackbox housing for future retrieval and examination and the secondary user (102) have access to the Blackbox data for the identification of any safety breaches. AI model running on the small form-factor GPU unit also classifies driver actions while driving and rates the overall safety quotient of the driver, logging into the Blackbox for future retrieval and analysis.

[0045] In an aspect, the present disclosure relates to a method for monitoring active safety of a vehicle. FIG. 3 illustrates an example method (300) for active vehicular safety, in accordance with an embodiment of the present disclosure. The system and other components referred to in the method (300) are

similar to the system (108) and components shown in FIG. 1. The method (300) includes the following steps. At step (302), the system (108) may receive, data comprising one or more first images, one or more second images, and one or more parameters, via one or more first image detection units (112), one or more second image detection units (112), and one or more sensors (114), at one or more predetermined periods by a data collection engine (210). At step 304, the system (108) detects, by a detecting and predicting engine (214), one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, reduction of load in the vehicle, distribution of load, and the like. At step 306, the system (108) may analyze, the data, using an Artificial Intelligence (AI) engine (212). At step 308, the system records the data received from the data collection engine (210) by a Blackbox housing and the overall safety performance of a primary user (110) is accessed based on the findings stored in the Blackbox housing for future retrieval and examination. The secondary user (102) has access to the Blackbox data for the identification of any safety breaches. At step 310, the system (108) may detect one or more risk factors, based on the analysis of the data from the Blackbox housing by a processor (202). At step 312, the system (108) alerts, by an alerting and reporting engine (216), secondary user(s) regarding risk factors, traffic violations, and the like.

[0046] In an embodiment, a pre-trained AI-model based on detailed video capture of the route traversed by a pilot vehicle driven by an expert driver is disclosed. The model captures each curve characteristics and the appropriate driver responses to create a benchmark for safe driving by other drivers. Such a system is able to operate without determining the location of the vehicle or external communication with any fixed or mobile infrastructure. It will alert the driver when a steep climb, curve or blind curve is approaching based on its pre-trained model.

[0047] In an exemplary embodiment, the computer devices in the present disclosure may be similar to the computer system as shown in FIG. 4. As shown in FIG. 4, the computer system (400) may include an external storage device (410), a bus (420), a main memory (430), a read-only memory (440), a mass storage device (450), a communication port(s) (460), and a processor (470). A person skilled in

the art will appreciate that the computer system (400) may include more than one processor and communication ports. The processor (470) may include various modules associated with embodiments of the present disclosure. The communication port(s) (460) may be any of an RS-232 port for use with a modem-based dialup connection, a 10/100 Ethernet port, a Gigabit or 10 Gigabit port using copper or fiber, a serial port, a parallel port, or other existing or future ports. The communication ports(s) (460) may be chosen depending on a network, such as a Local Area Network (LAN), Wide Area Network (WAN), or any network to which the computer system (400) connects.

[0048] In an embodiment, the main memory (430) may be Random Access Memory (RAM), or any other dynamic storage device commonly known in the art. The read-only memory (440) may be any static storage device(s) e.g., but not limited to, a Programmable Read Only Memory (PROM) chip for storing static information e.g., start-up or basic input/output system (BIOS) instructions for the processor (470). The mass storage device (450) may be any current or future mass storage solution, which can be used to store information and/or instructions. Exemplary mass storage solutions include, but are not limited to, Parallel Advanced Technology Attachment (PATA) or Serial Advanced Technology Attachment (SATA) hard disk drives or solid-state drives (internal or external, e.g., having Universal Serial Bus (USB) and/or Firewire interfaces).

[0049] In an embodiment, the bus (420) may communicatively couple the processor(s) (470) with the other memory, storage, and communication blocks. The bus (420) may be, e.g. a Peripheral Component Interconnect PCI) / PCI Extended (PCI-X) bus, Small Computer System Interface (SCSI), USB, or the like, for connecting expansion cards, drives, and other subsystems as well as other buses, such a front side bus (FSB), which connects the processor (470) to the computer system (400).

[0050] In another embodiment, operator and administrative interfaces, e.g., a display, keyboard, and cursor control device may also be coupled to the bus (420) to support direct operator interaction with the computer system (400). Other operator and administrative interfaces can be provided through network

connections connected through the communication port(s) (460). Components described above are meant only to exemplify various possibilities. In no way should the aforementioned exemplary computer system (400) limit the scope of the present disclosure.

[0051] While considerable emphasis has been placed herein on the preferred embodiments, it will be appreciated that many embodiments can be made and that many changes can be made in the preferred embodiments without departing from the principles of the disclosure. These and other changes in the preferred embodiments of the disclosure will be apparent to those skilled in the art from the disclosure herein, whereby it is to be distinctly understood that the foregoing descriptive matter is to be implemented merely as illustrative of the disclosure and not as a limitation.

ADVANTAGES OF THE PRESENT DESCRIPTION

[0052] The present disclosure provides a system and a method for monitoring active safety of a vehicle.

[0053] The present disclosure provides a system and a method for monitoring active safety of a vehicle which can also be used to assess a driver in terms of his/her driving skills, actions, and practices for safety, and conditions associated with a vehicle in real-time and/or over a period of time.

[0054] The present disclosure provides a system and a method for monitoring active safety of a vehicle that can be used in investigating safety violations, and performing analytics for developing interventions to promote vehicular safety by regulatory agencies and authorities.

We Claim:

1. A system (108) for monitoring safety of a vehicle, the system (108) comprising:

one or more first image detection units (112-1) configured with an interior of a vehicle to detect one or more first images of the interior of the vehicle, wherein the one or more first image detection units (112-1) are communicatively coupled to the system (108);

one or more second image detection units (112-2) configured with an exterior of a vehicle to detect one or more second images of one or more surroundings of the vehicle, wherein the one or more second image detection units (112-2) are communicatively coupled to the system (108);

one or more sensors (114) configured with the vehicle to detect one or more parameters associated with the vehicle and a movement of the vehicle to monitor for a vehicular safety, wherein the one or more sensors (114) are communicatively coupled to the system (108);

at least one processor (202); and

a memory (204) operatively coupled with the at least one processor (202), wherein said memory (204) stores executable instructions which when executed by the processor (202), causes the processor (202) to:

receive, data comprising the one or more first images, the one or more second images, and the one or more parameters associated with the vehicle, via the one or more first image detection units (112), the one or more second image detection units (112), and the one or more sensors (114), at one or more predetermined periods;

detect one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, reduction of load in the vehicle, distribution of load, and the like;

analyze, the data, using an Artificial Intelligence (AI) engine (212); wherein the data includes anticipating upcoming curves and advising the optimal speed for safe navigation;

detect, one or more risk factors, based on the analysis of the data; and

alert secondary user(s) regarding risk factors, traffic violations, and the like.

2. The system (108) as claimed in claim 1, wherein the one or more sensors (114) are preferably one or more pressure sensors to detect pressure associated with a plurality of wheels of the vehicle, one or more luggage compartments, and one or more rooftop luggage carriers, indicative of the weight distribution across the vehicle.

3. The system (108) as claimed in claim 1, wherein the processor (202) analyzes the one or more parameters to determine a total load configured with the vehicle, and a distribution of the total load with one or more locations of the vehicle.

4. The system (108) as claimed in claim 1, wherein the processor (202) analyzes one or more first images for determining a number of passengers within the vehicle.

5. The system (108) as claimed in claim 1, wherein the speed of the vehicle is determined by the processor embedded with an artificial intelligence engine using the rate of change of the video frames captured.

6. The system (108) as claimed in claim 1, wherein the processor (202) notifies one or more secondary users (102) via one or more computing devices (104) associated with the one or more secondary users (102) about one or more safety violations, based on the analysis of the data.

7. The system (108) as claimed in claim 1, wherein the processor (202) stores the data with one or more first images, the one or more second images, and one or more parameters within a database (220).

8. The system (108) as claimed in claim 1, wherein the processor (202) evaluates and quantifies the primary user (110) for the vehicular safety, based on the analysis of the data.

9. The system (108) as claimed in claim 1, wherein the system (108) includes a Blackbox housing to record all data obtained from the image detection units (112) and pressure sensors, wherein the driver's overall safety performance is accessed, with the findings stored in the Blackbox housing for future retrieval and examination and the secondary user (102) have access to the Blackbox data for the identification of any safety breaches.

10. A method (300) for monitoring active safety of a vehicle using the system (108), the method (300) comprising:

receiving (302), by a data collection engine (210), data comprising the one or more first images, the one or more second images, and the one or more parameters associated with the vehicle, via the one or more first image detection units (112), the one or more second image detection units (112), and the one or more sensors (114), at one or more predetermined periods;

detecting (304), by a detecting and predicting engine (214), one or more risk factors, and predict one or more safe parameters for the operation of the vehicle which may include speed, driver responses, reduction of load in the vehicle, distribution of load, and the like;

analyzing (306), the data, using an Artificial Intelligence (AI) engine (212); wherein the data includes anticipating upcoming curves and advising the optimal speed for safe navigation;

recording (308), by a Blackbox housing, data received from the data collection engine (210) and accessing the overall safety performance of a primary user (110) based on the findings stored in the Blackbox housing for future retrieval and examination; wherein the secondary user (102) have access to the Blackbox data for the identification of any safety breaches;

detecting (310), by a processor (202), one or more risk factors, based on the analysis of the data from the Blackbox housing; and

alerting (312), by an alerting and reporting engine (216), secondary user(s) regarding risk factors, traffic violations, and the like.

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ABSTRACT

SYSTEM FOR MONITORING ACTIVE SAFETY OF A VEHICLE AND METHOD THEREOF

The present disclosure provides a system and a method for monitoring active safety of a vehicle. The system (108) includes image detection unit(s) (112) configured with a vehicle to detect image(s) associated with an interior and exterior (surroundings) of the vehicle, and sensor(s) (114) configured with the vehicle to detect parameters associated with the vehicle and a movement of the vehicle to monitor for a vehicular safety. The system (108) receives, data including the image(s), and the parameter(s), via the image detection unit(s) (112) and the sensor(s) (114), at predetermined periods. The system (108) analyzes, the data, using an Artificial Intelligence (AI) engine, and detects, risk factors, based on the analysis of the data. The system (108) may predict driving parameter(s) based on the analysis of the data for safe driving.

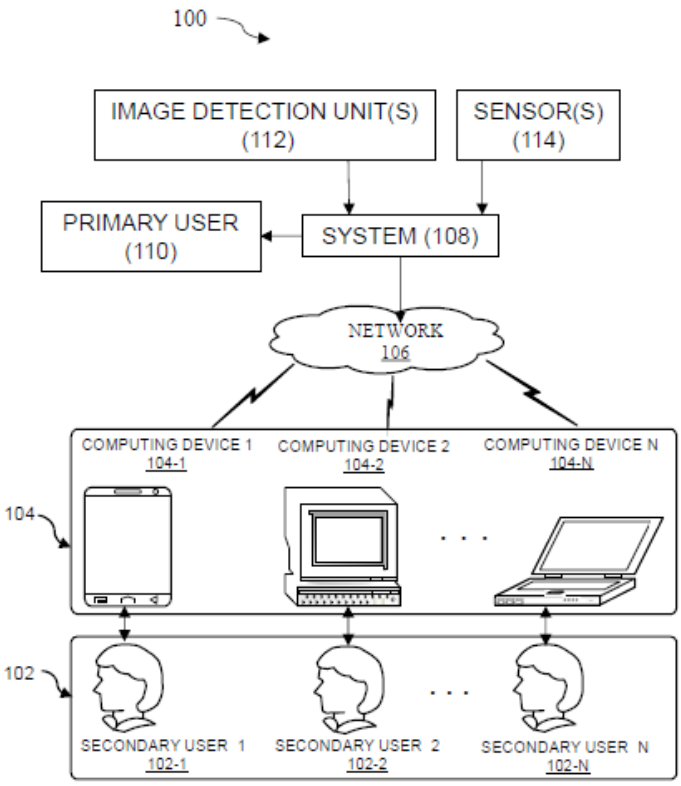


FIG. 1